



Still no evidence for an environmentally responsive epigenetic clock in an insect

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Insect diapause represents a remarkable evolutionary adaptation that enables these organisms to endure shifting environmental conditions by halting most biological functions, including developmental processes (1).

In a recent Brief Report (2), Foley et al. propose that larval diapause induced by altering light:dark photoperiods in the wasp *Nasonia vitripennis* slows down adult epigenetic aging. They argue that their findings are comparable to mammalian epigenetic clocks based on DNA methylation. However, their results do not support such a conclusion.

First, the authors constructed their DNA methylation clock by selecting age-associated sites and training a model on control samples. However, the model seems overfitted, lacking cross-validation, and the chosen CpG sites were based solely on all control samples, resulting in an apparent age correlation. When tested on diapause samples, the regression slope is lower. This is expected because any model trained on one dataset and applied to an external dataset will generally yield attenuated regression slopes, as the slope is proportional to the correlation between predicted and actual ages, which will be lower in the external set (3, 4). A more meticulous experiment would compare the model's performance on both external control and diapause samples or use larger, diverse lineage samples from field and lab settings.

Second, attempting to contextualize findings within the framework of development overlooks a key aspect of holometabolous insect development: complete metamorphosis. Most larval tissues are discarded, and new adult structures emerge from imaginal discs (5, 6), resulting in minimal cellular continuity in *Nasonia* development. This prompts questions about how larval epigenomic modifications influence adult epigenetics. The authors' claim depends entirely on adult samples, with diapause adults appearing epigenetically older by 2.8 d compared to controls, attributed to undefined epigenetic changes in larvae. However, there is no explanation for how larval modifications are transferred to adults. To support these assertions, it is essential to obtain larval and pupal samples. Furthermore, although survival rates

differ between diapause-derived and control adults, their predicted epigenetic ages overlap significantly, complicating the distinction between the groups.

Third, the authors utilized entire insect bodies, a methodology that would typically be deemed unacceptable in epigenomic studies involving aging in complex organisms. It is well established that various organs and tissues age at different rates within a single individual, and this asynchrony in aging has been observed at multiple cellular and physiological levels (7).

Fourth, the authors' use of low-depth whole-genome bisulfite sequencing might lead to an imbalanced representation of regions. Since only 27 CpGs were used to predict aging, higher sequencing depth is crucial for revealing tissue-type methylation patterns influenced by diapause (8).

Last, no assessment has been conducted on how genotype and epialleles influence methylation (9, 10). Therefore, the assertion that "epigenetic aging in this model is measurable and developmentally modifiable," along with the proposed cis-regulatory effects of gene body DNA methylation on the mTOR and insulin pathways, lacks substantial evidence.

While this report raises important questions for future research, it appears somewhat hastily compiled from a preliminary study that lacks several crucial experiments, which could bolster the credibility of the authors' claims.

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